Benoît Pasquier

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Summary

I am a researcher with expertise in multiple disciplines, including mathematics, computational science, oceanography, and finance. I have experience with running and diagnosing ocean and climate models, developing biogeochemistry models, parameter optimisation, and analysing large datasets. To support my scientific enquiries, I have developed several open-source scientific software packages. I have an excellent publication record and presented my work at local and international conferences. I am also passionate about education despite limited opportunities for teaching.

In the last few years, I have been investigating the ocean's response to global warming. In the future, I hope to leverage my skills in oceanography, climate modelling, and data analysis to develop innovative solutions and strategies that mitigate the impacts of climate change.

Professional Experience

Sep 24—Feb 25 **Contract Researcher** CSIRO, Hobart, Australia
Ocean transport matrices for investigating marine Carbon Dioxide Removal (mCDR).
Supervisor: Dr. Richard Matear.

Oct 21—Aug 24 **Research Associate** University of New South Wales, Sydney, Australia
Response of the ocean's carbon and oxygen cycles to climate change.
Supervisor: A/Prof. Mark Holzer.

Nov 19—Oct 21 **Postdoctoral Researcher** University of Southern California, Los Angeles, CA, USA
Global marine trace metals and isotopes modelling.
Supervisor: A/Prof. Seth John.

Sep 17—Sep 19 **Postdoctoral Research Scholar** University of California, Irvine, CA, USA
New tools for improving global biogeochemistry models.
Supervisors: Prof. François Primeau and Prof. J. Keith Moore.

Mar 17—Aug 17 **Casual Research Assistant** University of New South Wales, Sydney, Australia
Continued PhD work.
Supervisor: A/Prof. Mark Holzer.

Jun 16—Dec 16 **Mathematics Tutor** University of New South Wales, Sydney, Australia
Numerical Methods and Statistics, 2nd year.
Supervisor: Dr. Shev MacNamara.

May 11—Aug 12 **Proposal Engineer** Degrémont, Suez Environnement, Sydney, Australia
Tendering project management for design, construction, maintenance, and operational contracts. Business development, liaison with clients, advertising on company capabilities.

Jul 08—Jun 09 **Forex Trader Assistant** Société Générale Investment Banking, Paris, France
MASEF Internship in foreign exchange market (Forex). Research and software development in automated arbitrage using real-time high-frequency data.
Supervisors: Prof. Bruno Bouchard and Dr. Nicolas Boitout.

Apr 07—Jul 07 **Mathematics Research Intern** École Polytechnique, Palaiseau, France
École Polytechnique Speciality (Mathematics) Internship at the Laurent Schwartz Mathematics Centre (CMLS). Research review on the Witt ring of quadratic forms.
Supervisor: Prof. Jean Lannes.

Sep 04—Feb 05 **IT Intern** Bioforce, Lyon, France
Development of an ACCESS database to improve communication and management of Bioforce, which provides training and career advice in aid programs and logistics.

Jul 06—Jul 06 **Assembly Line Worker (Internship)** Mecaplast, Monaco
École Polytechnique Industrial Placement.

Education

2013—2017 **PhD in Applied Mathematics** University of New South Wales, Sydney, Australia
Thesis title: *The Ocean's Global Iron, Phosphorus, and Silicon Cycles: Inverse Modelling and Novel Diagnostics*.
Supervisor: A/Prof. Mark Holzer.

- Global Biogeochemical Cycles, Global Biological Pump
- Ecosystem Modelling & Biogenic Transport Modelling
- Green Functions Techniques (Path Densities, Flow Rates, Time Scales)
- Nonlinear Systems, Parameter Optimisation/Inverse Modelling
- Iron Control on the Global Biological Pump
- Southern Ocean Nutrient Trapping

2010 **MSc in Environmental Science** University of New South Wales, Sydney, Australia
Study of the nature of environmental problems and the methodology of their evaluation and management.

- Geophysical Fluid Dynamics
- Oceanography
- Project Management, Environmental Risk Management

2007—2008 **MSc in Finance Mathematics** Paris Dauphine + ENSAE ParisTech, Paris, France
MASEF (Mathematics of Insurance, Economics and Finance), Finance speciality.
Supervisor: Prof. Bruno Bouchard.
- Stochastic Calculus, Levy Processes with Jumps
- Stochastic Differential Equations
- Numerical Methods (Monte Carlo)

2004—2007 **MSc in Mathematics & Engineering** École Polytechnique, Palaiseau, France
Pure mathematics specialisation.
- Algebra, Arithmetic, Numerical Methods, Computer Science
- Differential Topology, Relativity
- Physics, Biology

2001—2004 **Preparatory Classes** Lycée Masséna, Nice, France
French Preparatory Classes, mathematics speciality.
- Linear Algebra, Topology, Numerical Methods, Computer Science
- Mechanics, Electromagnetism, Thermodynamics

Publications

I have published 16 peer-reviewed articles including 8 first-author works. My work has been cited 232 times, my h-index is 9, and my i10-index is 9 (Google Scholar).

16. Deoxygenation and Its Drivers Analyzed in Steady State for Perpetually Slower and Warmer Oceans
Benoît Pasquier, Mark Holzer, Matthew A. Chamberlain, Richard J. Matear, Nathaniel L. Bindoff
Journal of Geophysical Research: Oceans *129.9 (2024) e2024JC021043* DOI: 10.1029/2024JC021043

15. The Biological and Preformed Carbon Pumps in Perpetually Slower and Warmer Oceans
Benoît Pasquier, Mark Holzer, Matthew A. Chamberlain
Biogeosciences *21.14 (July 2024) pp. 3373–3400* DOI: 10.5194/bg-21-3373-2024

14. Biogeochemical Fluxes of Nickel in the Global Oceans Inferred From a Diagnostic Model
Seth G. John, Hengdi Liang, **Benoît Pasquier**, Mark Holzer, Sam Silva
Global Biogeochemical Cycles *38.5 (2024) e2023GB008018* DOI: 10.1029/2023GB008018

13. Atmospheric pCO2 Response to Stimulated Organic Carbon Export: Sensitivity Patterns and Timescales
Mark Holzer, Tim DeVries, **Benoît Pasquier**
Geophysical Research Letters *51.12 (2024) e2024GL108462* DOI: 10.1029/2024GL108462

12. Optimal Parameters for the Ocean's Nutrient, Carbon, and Oxygen Cycles Compensate for Circulation Biases but Replumb the Biological Pump
Benoît Pasquier, Mark Holzer, Matthew A. Chamberlain, Richard J. Matear, Nathaniel. L. Bindoff, François. W. Primeau
Biogeosciences *20.14 (2023) pp. 2985–3009* DOI: 10.5194/bg-20-2985-2023

11. The Biogeochemical Balance of Oceanic Nickel Cycling
Seth G. John, Rachel L. Kelly, Xiaopeng Bian, Feixue Fu, M. Isabel Smith, Nathan T. Lanning, Hengdi Liang, **Benoît Pasquier**, Emily A. Seelen, Mark Holzer, Laura Wasylenki, Tim M. Conway, Jessica N. Fitzsimmons, David A. Hutchins, Shun-Chung Yang
Nature Geoscience *15.11 (2022) pp. 906–912. Nature Publishing Group* DOI: 10.1038/s41561-022-01045-7

10. AIBECS.Jl: A Tool for Exploring Global Marine Biogeochemical Cycles.
Benoît Pasquier, François W. Primeau, Seth G. John
Journal of Open Source Software *7.69 (2022) p. 3814* DOI: 10.21105/joss.03814

9. GNOM v1.0: An Optimized Steady-State Model of the Modern Marine Neodymium Cycle
Benoît Pasquier, Sophia K. V. Hines, Hengdi Liang, Yingzhe Wu, Steven L. Goldstein, Seth G. John
Geoscientific Model Development *15.11 (2022) pp. 4625–4656* DOI: 10.5194/gmd-15-4625-2022

8. A New Metric of the Biological Carbon Pump: Number of Pump Passages and Its Control on Atmospheric pCO2
Mark Holzer, Eun Young Kwon, **Benoît Pasquier**
Global Biogeochemical Cycles *35.6 (2021) e2020GB006863* DOI: 10.1029/2020GB006863

7. Evaluating the benefits of Bayesian hierarchical methods for analyzing heterogeneous environmental datasets: a case study of marine organic carbon fluxes
Gregory L. Britten, Yara Mohajerani, Louis Primeau, Murat Aydin, Catherine Garcia, Wei-Lei Wang, **Benoît Pasquier**, B. B. Cael, François W. Primeau
Frontiers in Environmental Science *9 (2021) p. 28* DOI: 10.3389/fenvs.2021.491636

6. Perspective on identifying and characterizing the processes controlling iron speciation and residence time at the atmosphere-ocean interface
Nicholas Meskhidze, Christoph Völker, Hind A. Al-Abadleh, Katherine Barbeau, Matthieu Bressac, Clifton Buck, Randelle M. Bundy, Peter Croot, Yan Feng, Akinori Ito, Anne M. Johansen, William M. Landing, Jingqiu Mao, Stelios Myriokefalitakis, Daniel Ohnemus, **Benoît Pasquier**, Ying Ye
Marine Chemistry *217 (2019) p. 103704* DOI: 10.1016/j.marchem.2019.103704

5. Diatom Physiology Controls Silicic Acid Leakage in Response to Iron Fertilization
Mark Holzer, **Benoît Pasquier**, Timothy DeVries, Mark Brzezinski
Global Biogeochemical Cycles *33.12 (2019) pp. 1631–1653* DOI: 10.1029/2019GB006460

4. The number of past and future regenerations of iron in the ocean and its intrinsic fertilization efficiency
Benoît Pasquier, Mark Holzer
Biogeosciences *15.23 (2018) pp. 7177–7203* DOI: 10.5194/bg-15-7177-2018

3. Inverse-model estimates of the ocean's coupled phosphorus, silicon, and iron cycles
Benoît Pasquier, Mark Holzer
Biogeosciences *14.18 (2017) pp. 4125–4159* DOI: 10.5194/bg-14-4125-2017

2. The age of iron and iron source attribution in the ocean
Mark Holzer, Marina Frants, **Benoît Pasquier**
Global Biogeochemical Cycles *30.10 (2016) pp. 1454–1474* DOI: 10.1002/2016GB005418

1. The plumbing of the global biological pump: Efficiency control through leaks, pathways, and time scales
Benoît Pasquier, Mark Holzer

Journal of Geophysical Research: Oceans *121.8 (2016) pp. 6367–6388* DOI: 10.1002/2016JC011821

Talks and Posters

I have contributed to at least 28 conference talks and posters, including 18 as a first author.

28. What Sets Neodymium Distributions in the Ocean and How Does This Impact its Utility as a Paleoceanographic Proxy?
Sophia K. V. Hines, Paige M. Wise, **Benoît Pasquier**, Seth G. John
AGU Fall Meeting, 2024, Washington, DC, USA

27. Shallow vs Deep: Identifying the magnitude and locations of marine Nd sources with GNOM
Paige M. Wise, **Benoît Pasquier**, Seth G. John, Sophia K. V. Hines
Goldschmidt Conference, 2024, Chicago, IL, USA

26. The Ocean's Carbon and Oxygen Cycles in Future Steady-State Climate Scenarios
Benoît Pasquier, Mark Holzer, Matthew A. Chamberlain, Richard J. Matear, Nathaniel L. Bindoff, François W. Primeau
Ocean Sciences Meeting, 2024, New Orleans, Louisiana, USA

25. Optimal parameters for the ocean's nutrient, carbon, and oxygen cycles compensate for circulation biases but replumb the biological pump
Benoît Pasquier, Mark Holzer, Matthew A. Chamberlain, Richard J. Matear, Nathaniel L. Bindoff, François W. Primeau
AMOS National Conference, 2024, Canberra, Australia

24. Quantifying the Dominant Fluxes of the Modern Marine Neodymium Cycle through Model Optimization
Paige M. Wise, Lowell Stott, Seth G. John, **Benoît Pasquier**
AGU Fall Meeting, 2023, San Francisco, USA

23. Global distribution of nickel sources and sinks from a diagnostic model
Seth G. John, Hengdi Liang, **Benoît Pasquier**, Mark Holzer, Sam Silva
Goldschmidt Conference, 2023, Lyon, France

22. PCO2: A simple biogeochemistry model embedded in a simple ocean circulation model in matrix form
Benoît Pasquier
UNSW Ocean Research Carnival, 2023, UNSW, Sydney, Australia

21. The Biogeochemical Balance that Controls Oceanic Nickel Cycling in the Modern and Past Oceans
Seth G. John, Rachel Kelly, Xiaopeng Bian, Shun-Chung Yang, Feixue Fu, Magdalene I. Smith, Nathan Lanning, Hengdi Liang, **Benoît Pasquier**, Emily Seelen, Mark Holzer, Tim M. Conway, Jessica N. Fitzsimmons, David Hutchins
Goldschmidt Conference, 2022, Honolulu, Hawaii, USA

20. Modeling Marine Ecosystems At Multiple Scales Using Julia
Gaël Forget, **Benoît Pasquier**, Zhen Wu
JuliaCon, 2021, Online

19. Global Controls on the Distribution of Nickel in the Oceans
Seth G. John, Shun-Chung Yang, Xiaopeng Bian, Rachel Kelly, David Hutchins, Feixue Fu, Hengdi Liang, **Benoît Pasquier**, Emily Seelen
Goldschmidt Conference, 2020, Online

18. Metals, Models, and Isotopes: Insights into the Biogeochemical Cycling of Trace Nutrients in the Ocean
Seth G. John, Tim M. Conway, Tom Weber, Tim DeVries, Alessandro Tagliabue, Hengdi Liang, **Benoît Pasquier**
Goldschmidt Conference, 2020, Online

17. Julia users and tools for oceanography
Gaël Forget, **Benoît Pasquier**, Alexander Barth, Milan Klöwer, Ali Ramadan, Gregory L. Wagner, Constantinou Navid
Ocean Sciences Meeting, 2020, San Diego Convention Center, San Diego, California, USA

16. AIBECS.jl: the ideal tool for marine biogeochemistry modelling
Benoît Pasquier, François Primeau
Ocean Sciences Meeting, 2020, San Diego Convention Center, San Diego, California, USA

15. F-1 algorithm: Efficient differentiation through large steady-state problems
Benoît Pasquier, François Primeau
Applied Maths Seminar, 2019, School of Mathematics and Statistics, UNSW, Australia

14. Introducing AIBECS.jl, a Julia package for creating global marine biogeochemistry models
Benoît Pasquier, François Primeau, J. Keith Moore
CCRC Seminars, 2019, Climate Change Research Centre (CCRC), UNSW, Australia

13. Southern Ocean silicic-acid leakage: Sensitivity to diatom physiology and its Si isotope signature analyzed in an inverse model
Mark Holzer, **Benoît Pasquier**, Tim DeVries, Mark Brzezinski
AMOS National Conference, 2019, Darwin, Australia

12. The number of past and future regenerations of iron in the ocean and its intrinsic fertilization efficiency
Benoît Pasquier, Mark Holzer
Michael Follows Group Meeting, 2019, MIT, USA

11. Developing a new, open-source, user-friendly, fast, modular, global marine biogeochemistry model (in Julia)
Benoît Pasquier
Sack-lunch seminar, 2019, MIT, USA

10. Offline parameter optimization for global marine biogeochemical models
Benoît Pasquier
François Primeau Group Meeting, 2018, University of California, Irvine, USA

9. Inverse-model estimates of the ocean's coupled phosphorus, silicon, and iron cycles.
Benoît Pasquier, Mark Holzer
Ocean Sciences Meeting, 2018, Portland, Oregon, USA

8. The efficiency of different iron sources in supporting the ocean's global biological pump
Benoît Pasquier, Mark Holzer
Half-baked seminar, Department of Earth System Science, 2017, University of California, Irvine, USA
7. Response of the biological pump to perturbations in the iron supply: Global teleconnections diagnosed using an inverse model of the coupled phosphorus-silicon-iron nutrient cycles
Benoît Pasquier, Mark Holzer
AMOS National Conference, 2017, Canberra, Australia
6. Exploring iron control on global productivity: "FePSi", an inverse model of the ocean's coupled phosphate, silicon and iron cycles
Benoît Pasquier, Mark Holzer
Postgrad Conference, 2016, Sydney, Australia
5. Iron Source Attribution and the Age of Dissolved Iron in the Ocean
Mark Holzer, **Benoît Pasquier**, Marina Frants
Ocean Sciences Meeting, 2016, New Orleans, Louisiana, USA
4. Iron control on global productivity: an efficient inverse model of the ocean's coupled phosphate, silicon, and iron cycles
Benoît Pasquier, Mark Holzer
Ocean Sciences Meeting, 2016, New Orleans, Louisiana, USA
3. The plumbing of the global biological pump
Benoît Pasquier, Mark Holzer
AMOS National Conference, 2015, Brisbane, Australia
2. An efficient inverse model of the ocean's coupled nutrient cycles
Benoît Pasquier, Mark Holzer
Postgrad Conference, 2015, Sydney, Australia
1. Plumbing of the biological pump
Benoît Pasquier, Mark Holzer
Postgrad Conference, 2014, Sydney, Australia

Honors and Awards

2021 **Outstanding Review** — AGU Global Biogeochemical Cycles editors
https://eos.org/agu-news/in-appreciation-of-agus-outstanding-reviewers-of-2021

2015 **Scholarship** — Cuomo Foundation, Monaco

2014 **Scholarship** — Frères Louis et Max Principale Foundation, Monaco

2014 - 2016 **Scholarship** — Monaco Government, Monaco
Higher studies scholarship

2013 **Scholarship** — Monaco Government, Monaco
H.S.H. The Prince Albert II Exceptional Scholarship

2013 - 2016 **Scholarship** Monaco Scientific Centre, Monaco

2013 - 2016 **Tuition Fee Scholarship** Graduate Research School, UNSW, Sydney, Australia

2004 - 2008 **Scholarship** Monaco Government, Monaco
Higher studies scholarship

Skills

Programming		Languages	
Julia / MATLAB	Expert	French / English	Fluent
Python / LaTeX	Advanced	Italian	Intermediate
HPC / Shell scripting / HTML	Competent	Japanese	Novice
Mathematica / Maple / SageMath	Competent		
FORTRAN / C++ / Ruby / R	Out of practice		

Open-source scientific software contributions

Owner OceanTransportMatrixBuilder.jl https://github.com/TMIP-code/OceanTransportMatrixBuilder.jl
A Julia package to build ocean transport matrices from CMIP model output.

Owner AIBECS.jl https://github.com/JuliaOcean/AIBECS.jl
The ideal tool for exploring global marine biogeochemical cycles.

Owner GNOM https://github.com/MTEL-USC/GNOM
An optimized steady-state model of the modern global marine neodymium cycle.

Owner F1Method.jl https://github.com/briochemc/F1Method.jl
Efficient quasi-auto-differentiation of an objective function defined implicitly by the solution of a steady-state problem.

Collaborator UnitfulRecipes.jl https://github.com/briochemc/UnitfulRecipes.jl
Plotting data with units seamlessly in Julia.

Owner Inpaintings.jl https://github.com/briochemc/Inpaintings.jl
Julia version of MATLAB's inpaint_nans.

Owner WorldOceanAtlasTools.jl https://github.com/briochemc/WorldOceanAtlasTools.jl
Downloading and using data from the World Ocean Atlas (WOA) database.

Owner OceanographyCruises.jl https://github.com/briochemc/OceanographyCruises.jl
An interface for dealing with oceanographic cruises data.

Contributor latexdiff https://github.com/ftilmann/latexdiff
Compares two latex files and marks up significant differences between them.

Contributor **YAXArrays.jl** https://github.com/JuliaDataCubes/YAXArrays.jl
Yet Another XArray-like Julia package.

Contributor **cmap** https://github.com/tsipkens/cmap
Perceptually uniform colormaps for MATLAB, compiled from multiple sources.

Contributor **Makie.jl** https://github.com/MakieOrg/Makie.jl
Interactive data visualisations and plotting in Julia.

Contributor **offsetaxis-pkg** https://github.com/kakearney/offsetaxis-pkg
OFFSETAXIS Add an x- or y-axis offset from the plotted axis area (MATLAB).

Contributor **GeoStats.jl** https://github.com/JuliaEarth/GeoStats.jl
Comprehensive framework for geostatistics (or spatial statistics).

Owner **OceanGrids.jl** https://github.com/briochemc/OceanGrids.jl
Standard format of grids for AIBECS.

Owner **OceanBasins.jl** https://github.com/briochemc/OceanBasins.jl
Programmatically determine which ocean basin a (lat,lon) coordinate is in.

Owner **GEOTRACES.jl** https://github.com/briochemc/GEOTRACES.jl
A package for reading and using GEOTRACES data in Julia.

Collaborator **HyperDualNumbers.jl** https://github.com/JuliaDiff/HyperDualNumbers.jl
Julia implementation of HyperDualNumbers.

Owner **DualMatrixTools.jl** https://github.com/briochemc/DualMatrixTools.jl
Efficiently solve dual-valued linear systems.

Owner **HyperDualMatrixTools.jl** https://github.com/briochemc/HyperDualMatrixTools.jl
Efficiently solve hyperdual-valued linear systems.

Owner **BlockDiagonalFactors.jl** https://github.com/briochemc/BlockDiagonalFactors.jl
Efficiently solve linear block-diagonal systems with repeated blocks.

Contributor **Plots.jl** https://github.com/JuliaPlots/Plots.jl
Powerful convenience for Julia visualisations and data analysis.

Contributor **Unitful.jl** https://github.com/PainterQubits/Unitful.jl
Julia package for physical units.

Contributor **UnitfulMoles.jl** https://github.com/briochemc/UnitfulMoles.jl
A set of predefined conventional elemental mol units.

Contributor **Distributions.jl** https://github.com/JuliaStats/Distributions.jl
A Julia package for probability distributions and associated functions.

Contributor **DiffEqBase.jl** https://github.com/SciML/DiffEqBase.jl
DiffEqBase.jl is a component package in the DiffEq ecosystem.

Contributor **SciMLBase.jl** https://github.com/SciML/SciMLBase.jl
The Base interface of the SciML ecosystem.

Contributor **DiffEqOperators.jl** https://github.com/SciML/DiffEqOperators.jl
Linear operators for discretizations of differential equations and scientific machine learning (SciML).

Contributor **Interpolations.jl** https://github.com/JuliaMath/Interpolations.jl
Fast, continuous interpolation of discrete datasets in Julia.

Contributor **RecipesBase.jl** https://github.com/JuliaPlots/RecipesBase.jl
Base package for defining transformation recipes on user types for Plots.jl

Contributor **CMAP.jl** https://github.com/simonscmap/CMAP.jl
Simons CMAP Julia client.

Contributor **InverseDistanceWeighting.jl** https://github.com/juliohm/InverseDistanceWeighting.jl
Inverse distance estimation solver for the GeoStats.jl framework.

Owner **Earth2014.jl** https://github.com/briochemc/Earth2014.jl
Download topographic data for the globe.

Professional References

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